30.9 A 147TOPS/W, 250TOPS/mm², Fully Synthesizable, Digital Compute-in-Memory Accelerator Supporting INT8×INT8 with Zero-Point Quantization in Intel 18A Technology

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Abstract

A fully synthesizable DCiM on-die accelerator with 128 input channels, 32 output channels, and 2 weight sets, supporting zero-point-quantized INT8×INT8 computation, is fabricated in Intel 18A technology and occupies 0.0856mm². The DCiM accelerator operates at 2.62GHz for a supply voltage of 1.1V, and 25°C, achieving a peak area-efficiency of 250TOPS/mm². Robust operation of DCiM is maintained down to 400mV, 25°C, where it delivers a peak energy-efficiency of 147TOPS/W at 25% input activity, with a power consumption of 6.1mW.

Artificial Intelligence (AI) hardware accelerators continue to drive advances in computer vision, speech recognition, large language models (LLMs), and image and video processing. Compute-in-memory (CiM) architectures reduce data-movement overhead by integrating distributed memory and compute with wide adder-trees, enabling highly parallel computation. Digital compute-in-memory (DCiM) provides deterministic accuracy, with process and voltage scalability along with immunity to parametric variation, while a fully synthesizable design enables portability across technology nodes, and simplifies integration using standard design tools and cell libraries, promoting broader adoption. This work presents a fully synthesizable DCiM on-die accelerator featuring 128 input channels, 32 output channels, 2 weight sets, high-density multibit interruptible 12T-latch-based memory storage, radix-4 stationary weight encoded Booth computation, a unique unsigned-to-signed mapping scheme, and support for zero-point-quantized INT8×INT8 data format. Fabricated in Intel 18A technology [1] (Fig. 30.9.7), the DCiM accelerator operates at 2.62GHz measured at maximum supply voltage of 1.1V and 25°C, occupying a compact area of 0.0856mm² while achieving: (i) peak area efficiency of 250TOPS/mm² and throughput of 21.5TOPS, at 1.1V and 25°C, (ii) robust functionality of DCiM measured down to 400mV at 25°C with peak energy efficiency of 147TOPS/W and power consumption of 6.1mW, at 25% input activity, (iii) energy efficiency scalable to 254TOPS/W at 3.5mW for 10% input activity for supply voltage of 400mV and 25°C, (iv) unique unsigned-to-signed mapping and zero-point expansion with pre-computing saving 20% power/17% area, (v) Booth encoding on stationary weights with 28% reduction in compute power, and (vi) use of high-density multibit latch bitcell resulting in 20% reduction in weight-storage area.

The DCiM accelerator organization consists of 128 input channels and 32 output channels, supporting zero-point quantized 8b unsigned/signed activations and weights (Fig. 30.9.1). Zero-point quantization improves computation accuracy by quantizing more precisely for data not symmetrically centered around zero. However, this requires addition of an origin shift constant from activations and weights, resulting in an effective signed 9b computation. Pre-computing both an unsigned-to-signed bias and zero-point expansion converts the datapath to signed 8b×8b. The converted signed 8b weights are Booth encoded as four 3b Booth selects (single, neg, pos). These Booth selects for a compute column of 128 weights are written every cycle, completing the entire weight-set write in 32 cycles. A double buffered scheme stores two weight sets per compute column. Computations are performed on the selected weight set, which is read through a static 2:1 multiplexor only at the beginning of each compute operation for reduced read power. The second weight location can be updated in the background, eliminating any update stalls when the latency of compute operation is longer than writing a new set of weights. Balanced optimization of throughput and power results in four pipeline stages to complete 128-way MAC. The unsigned-to-signed conversion with zero-point offset is performed in the first pipeline stage while the next two stages complete the 32-way bank results. The converted signed 8b activations from the first stage are broadcast across all compute columns, generating four partial products per multiplication. Partial products are added using 32-way carry-save adder trees, first with addition across multipliers, followed by merging the shifted partial-products sum. Finally, the outputs from four banks are summed to complete 128-way MAC in the last pipeline stage.

When optimizing for power and area, Booth multipliers offer unique benefits when applied to DCiM datapaths. Radix-4 Booth encoding reduces partial products by half but requires one multiplier input to be Booth encoded. Therefore, this encoder overhead can limit the power and area savings of Booth multipliers, especially for low data bit widths. The DCiM dataflow is unique, because weight inputs are stored and remain stationary for multiple compute cycles, while activation inputs are broadcast across compute columns. Booth encoding can be applied to either the broadcast activation or the stored weights. The 8b activation can be encoded once in the periphery, with 12b Booth selects broadcast across the compute columns. This amortizes the encoder area and power across all 32 compute columns but increases the number of input bits switching every cycle from 8 to 12, thereby increasing compute power. Alternatively, pre-encoding 8b weights and storing 12b Booth selects eliminates encoder power during compute but increases storage area (Fig. 30.9.2). Since weights are stored and written per column, moving the weight encoder to the memory write path enables reuse and reduces encoder area by 32× due to lower throughput requirements. Booth S-bits, which only depend on multipliers (weights), are pre-computed and stored in the weight memory for later addition with the final sum. Furthermore, encoding of the weights preserves typically low MSB activity of activation inputs into the adder-tree. Overall, the implemented weight encoding design results in only 7% increase in area, while enabling 28% lower compute power with stationary weights, compared to activation encoding. Double buffered weights are stored using interruptible 12T multibit latch standard cells. This enables robust low supply voltage write operation and memory read through static 2:1 multiplexor followed by Booth selectors. A modified Booth encoding is designed with single/neg/pos instead of single/double/neg select signals, which encodes ±0 by forcing both neg/pos signal to either 0 or 1. This modified Booth encoding results in up to 12% lower Booth selector area.

Radix-4 Booth multiplication with 8b unsigned/signed activations and weights results in five partial products. Five carry-save adder trees sum these partial products in groups with the same alignment across multipliers. Zero-point quantization requires addition of a constant, converting these inputs to 9b signed integers. The radix-4 Booth with 9b signed multiplication would not only generate 5 partial products but would also increase their width from 9b to 10b (Fig. 30.9.3). Instead, a unique unsigned-to-signed mapping converts the 8b unsigned integer to 8b signed integer by subtracting a bias of 128. This conversion can be done by conditionally inverting the MSB based on input format. Likewise, the zero-point constant must also be added from every activation and weight. The bias and zero-point constant are combined and added to the mapped 8b signed integer, representing final activation and weights with zero-point quantization. The DCiM sum of product computation is represented by the equation ($\Sigma_N A_Z W_Z$) shown in Fig. 30.9.2. Expanding this equation splits each 9b×9b signed multiplication ($\Sigma_N A_Z W_Z$) into an 8b×8b signed multiplication ($\Sigma_N A_S W_S$) plus three extra terms. In the DCiM dataflow the activation inputs are shared across multiple compute columns. The weight inputs are stationary (stored in memory) and constant for multiple cycles. Therefore, the compute cost of the three extra terms is amortized to reduce area and power. The sum of the activations ($\Sigma_N A_S$) multiplied with the constant ($C+Z_W$) is computed once every cycle and shared across multiple compute columns. The sum of weights ($\Sigma_N W_S$) multiplied with the constant ($C+Z_A$) is precomputed and stored in memory during weight writes, and therefore does not consume power during compute operations. Finally, the third term $N(C+Z_W)(C+Z_A)$ is a fixed constant. All three terms are added at the end of the compute datapath. This intrinsic zero-point compute with unsigned-signed mapping within the DCiM dataflow reduces the 9b×9b signed multiplication to an 8b×8b signed multiplication. This reduces the partial product width by 10% and decreases the number of partial products from 5 to 4, resulting in only 4 adder trees to complete the MAC operation. This technique lowers overall DCiM power by 17% and area by 21%.

DCiM area and power breakdown are shown in Fig. 30.9.4. The storage latch and read multiplexor circuits contribute 21% of the area, even though they consume only 11% of total write power. However, the write data driver and decoder contribute 56% of write power but occupy only 3% of area. The Booth encoder along with S-bit and zero-point pre-computation contribute only 16% of write power and 2% of total area, because of sharing across 32 columns. Double buffering allows concurrent update and compute. When compute latency is higher than writing a new set of weights, the DCiM becomes compute power bound, leading to maximum energy-efficiency. Booth selectors and adder trees, which contribute 74% of the DCiM area, reach 98% of the total compute power. The design is implemented with five different latch types ranging from single-bit to high-density multibit with two cycle compute-after-write timing constraints. This optimized timing achieves 95.5% high-density multibit insertion ratio, resulting in 20% weight storage area reduction.

The DCiM is synthesized using Intel 18A standard cell library, occupying total accelerator area of 0.0856mm². The DCiM implements 128 input channels, 32 output channels and 2 weight sets supporting 8b unsigned/signed zero-point quantized activations and weights. Weights are stored using interruptible 12T latch standard cells for robust low supply voltage write operation. The fabricated test-chip demonstrates a maximum throughput of 21.5TOPS at 2.62GHz, consuming 1.23W, with peak area efficiency of 250TOPS/mm² (measured at 1.1V and 25°C) (Fig. 30.9.5). The DCiM achieves peak energy efficiency of 147TOPS/W for input activity of 25%, weight 1=50%, consuming 6.1mW (measured at 400mV and 25°C). Energy efficiency scales to 254TOPS/W at 10% input activity, consuming 3.5mW. The double buffered scheme enables MAC computations on the selected weight set, while the

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second weight set is updated in the background for 32 cycles. As the number of total compute cycles increases, the write power contribution reduces to 1% and the DCiM becomes compute power bound reaching peak energy efficiency. Figure 30.9.6 compares the proposed work with recent state-of-the-art digital CiMs, showing best-in-class area efficiency (TOPS/mm²) and energy efficiency (TOPS/W). Figure 30.9.7 presents the photograph of the test-chip, its die dimensions, and a performance summary table.

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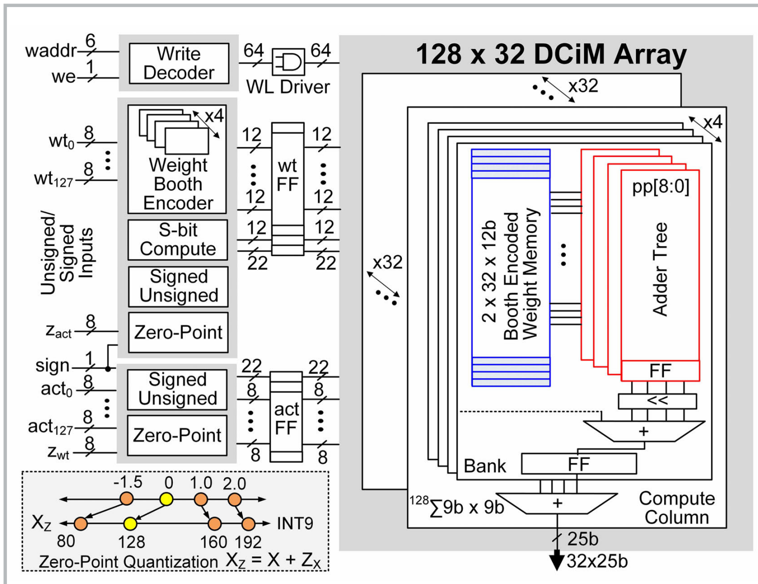


Figure 30.9.1: DCiM accelerator organization supporting zero-point quantized 8b unsigned/ signed activation and weights.

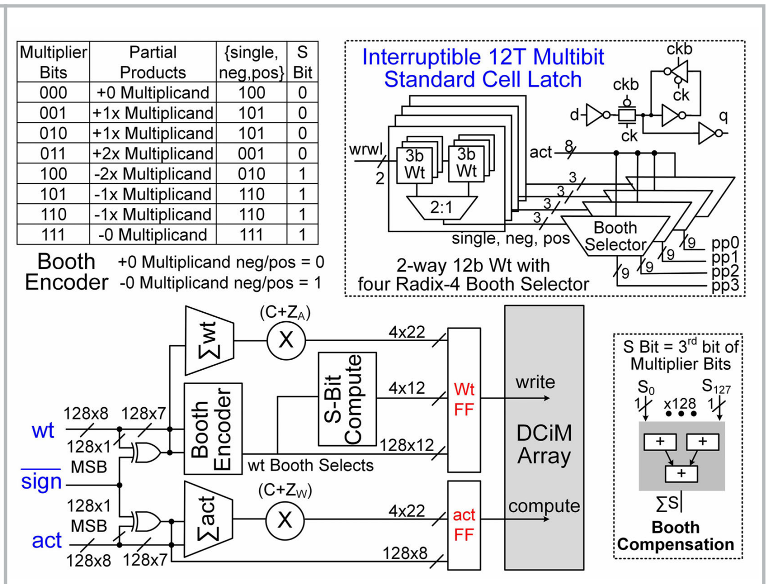
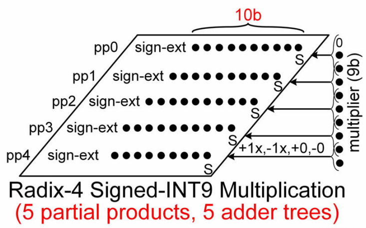
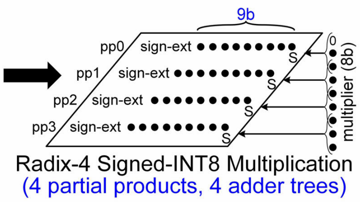


Figure 30.9.2: Radix-4 Booth encoded weights, interruptible 12T multibit latch storage with static 2:1 multiplexor read, S-bit compute, and modified Booth encoder.





Unsigned bin(dec)	Signed + 2^{m-1}(bias) bin(dec) + 2^{m-1}
00000000 (+0)	10000000 (-128) + 128
00000001 (+1)	10000001 (-127) + 128
⋮ ⋮ ⋮ Invert MSB	⋮ ⋮ ⋮
10000000 (+128)	00000000 (+0) + 128
⋮ ⋮ ⋮	⋮ ⋮ ⋮
11111110 (+254)	01111110 (+126) + 128
11111111 (+255)	01111111 (+127) + 128

Unsigned-INT8 to Signed-INT8 Mapping by adding bias (m=8)

Zero-Point Equation

$A_Z = A + Z_A$

$W_Z = W + Z_W$

signed INT9 — unsigned/ signed INT8 (5 adder trees)

with Unsigned-to-Signed

$A_Z = A_S + (C + Z_A)$

$W_Z = W_S + (C + Z_W)$

signed INT9 — signed INT8 — $C = 0$ if sign, $= 128$ else

$$\sum_N A_Z W_Z = \sum_N A_S W_S + (C+Z_W)\sum_N A_S + (C+Z_A)\sum_N W_S + N(C+Z_W)(C+Z_A)$$

Sum of product — signed-INT8 compute — computed once/cycle — pre-computed during weight write and stored — constant

**N* = number of Input Channel

Figure 30.9.3: Intrinsic zero-point compute with unsigned-signed mapping within the DCiM dataflow.

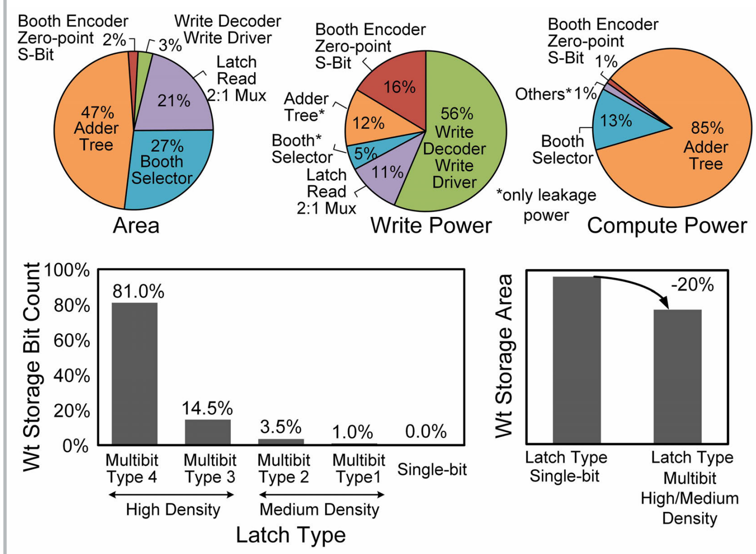


Figure 30.9.4: Area/power breakdown and weight storage area savings.

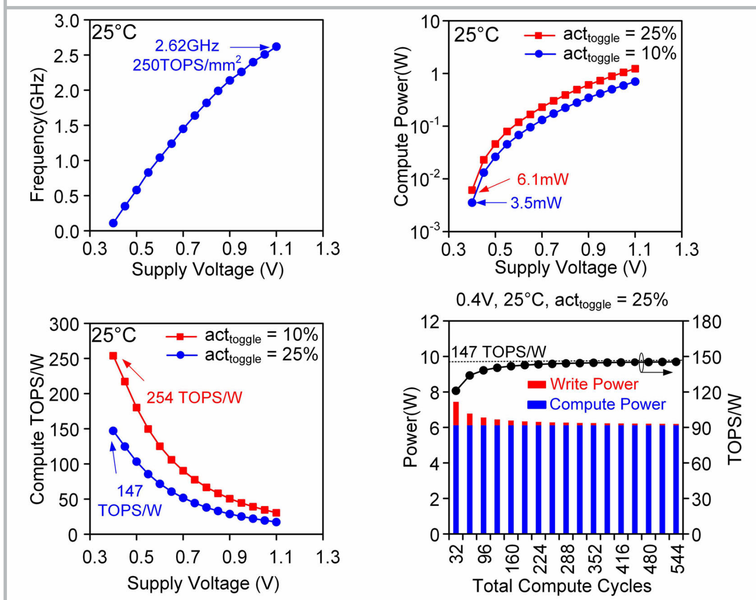
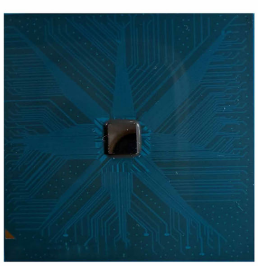


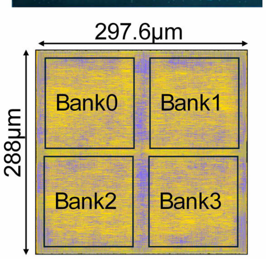
Figure 30.9.5: Frequency, power, and energy efficiency measurements in Intel 18A technology.

	ISSCC'23[2]	ISSCC'23 [3]	ISSCC'24 [4]	VLSI'25 [5]	This work
Technology	4nm	18nm	3nm	3nm	Intel 18A
MAC Operation	Digital	Digital	Digital	Digital	Digital
BitCell Type	8T	8T	6T	12T	12T
Macro Area	0.0172mm^2	0.0656mm^2	0.0167mm^2	0.0175mm^2	0.0856mm^2
Voltage	0.32-1.1V	0.525-1.0V	0.36-1.1V	0.37-1.1V	0.4-1.1V
Input Channel	144	32	72	64	128
Output Channel	16	64	4	8	32
Weight Set	2	32	18	18	2
Format	signed INT8/INT12 bit-serial	unsigned/ signed INT4	signed INT12	unsigned/ signed INT8	unsigned/signed INT8 zero-point INT9
Simultaneous Mac + Write	Yes	Yes	Yes	Yes	Yes
TOPS/mm^2	*49.9@0.9V	*3.4@1.0V	*124@0.9V	90.2@1.1V	250@1.1V
TOPS/W act_{toggle}=10%	--	--	*121@0.5V	124.6@0.5V	254@0.4V 180@0.5V
TOPS/W act_{toggle}=25%	*59.4@0.5V	*14.5@0.525V	*73.2@0.5V	70.4@0.5V	147@0.4V 104@0.5V

(*) Normalized to INT8

Figure 30.9.6: Power, performance, area summary, and comparison to prior work.





Technology	Intel 18A
Bitcell	Interruptible 12T multibit standard cell latch
DCiM Configuration	128 input channels 32 output channels 2 weight sets
Format	Unsigned/signed INT8, Zero-point INT9
Supply Voltage	0.4V-1.1V
Max Frequency	2.62GHz @1.1V
Macro Area	0.0856mm^2
TOPS/mm^2	250 @1.1V
TOPS/W	254 @0.4V (act_{toggle} =10%) 147 @0.4V (act_{toggle} =25%) 180 @0.5V (act_{toggle} =10%) 104 @0.5V (act_{toggle} =25%)

Figure 30.9.7: Packaged Intel 18A test-chip photograph, die dimensions, and performance summary table.

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